02 Conversion

Tuesday, July 21, 2026

9:12 PM

Why are we learning this?

In the previous lesson, we derived mole balances for CSTRs, batch reactors, and PFRs.

We could solve those balances, but the math gets a bit tricky. We will spend time practicing with these more complicated solutions later in the class. For now, we are going to simplify the mole balances.

To do this, we introduce **conversion**, one of the most important variables in reactor design. Conversion allows us to describe reactor performance using a single variable that directly answers the question: How much of the limiting reactant has been consumed? By the end of today's lesson, you should understand:

1. What conversion means physically.
2. How conversion differs in flow reactors and batch reactors.
3. How to write mole balances in terms of conversion.
4. Why conversion can reduce a multidimensional problem to a single unknown.
5. Why conversion is the foundation of many of the reactor design equations we will develop later.

Course Roadmap

Introduction to CRE
↓
General Balance Equation
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
►Reaction Progress◄
(Conversion)

↓

Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)

↓

Rate Laws
(Kinetics)

↓

Equilibrium
(Reversible Reactions)

↓

Equilibrium Conversion
(Thermodynamic Limitations)

↓

One Reaction: Solving in terms of Reaction Progress

↓

One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)

↓

Catalytic Reactions

↓

Numerical Methods
(Python)

↓

Energy Balances

↓

One Reaction: Non-Isothermal Reactor Design

↓

Multiple Reactions

↓

Reactor Safety

↓

Challenging the Well-Mixed Assumption
(Residence Time Distributions)

↓

Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Today's Location

Conversion

Today we connect reactor balances to reactor performance.

Big Ideas

Conversion answers a simple engineering question: What fraction of the limiting reactant has reacted? Rather than tracking the number of moles remaining or the change in the molar flowrate, we can instead track how much reaction has occurred. This transforms a complicated problem into:

One equation and one unknown.

In this class, we denote conversion with the letter X

The definition of conversion is

$$X = \frac{\text{amount reacted}}{\text{amount fed (flow reactor) or present initially (batch reactor)}}$$

What Does Conversion Measure?

Conversion is a measure of reactor performance. It tells us what fraction of the limiting reactant has been consumed by reaction.

Examples:

Conversion	Meaning
$X = 0$	No reaction has occurred
$X = 0.25$	25% of the limiting reactant has reacted
$X = 0.80$	80% of the limiting reactant has reacted
$X = 1$	Complete consumption of the limiting reactant

Check Yourself

Is high conversion always desirable?

(We'll later discover situations involving equilibrium, selectivity, and heat effects where the answer is not necessarily yes.)

Mathematically, X = $$X_A = \frac{F_{AO} - F_A}{F_{AO}}$$

in a flow reactor

And X = $$X_A = \frac{N_{AO} - N_A}{N_{AO}}$$

in a batch reactor

Why Are There Two Definitions of Conversion?

The physical idea is always the same:

Conversion = fraction of the limiting reactant that has reacted.

The denominator changes because the reference point changes.

Flow Reactor

We compare against the amount entering the reactor.

$$X = \frac{\text{amount reacted}}{\text{amount fed}}$$

Batch Reactor

We compare against the amount initially present in the reactor.

$$X = \frac{\text{amount reacted}}{\text{amount initially present}}$$

The concept is identical. Only the bookkeeping changes.

For the special case where there is only 1 reaction and the mole balances are only influenced by the reaction (and not by moles being added or removed by other means such as diffusion through a membrane), it is useful to *solve* the mole balance in terms of X, as this reduces a multidimensional problem to "1 equation and 1 unknown".

Let's convert the mole balances for the 3 common reactors so that they are in terms of X instead of

molar flowrate or amount of moles
(flow reactors) (Batch reactor)

CSTR:

$V = \frac{F_{A0}-F_A}{-r_A}$ and $X = \frac{F_{A0}-F_A}{F_{A0}}$.

Step 1: Write F_A in terms of X: $F_A = F_{A0}(1-X)$

Step 2: Substitute into the mole balance:

$$V = \frac{F_{AO} - F_A}{-r_A} = \frac{F_{AO} - F_{AO}(1-X)}{-r_A}$$

$$\boxed{V = \frac{F_{AO}X}{-r_A}}$$

if I use this form
I need to write
-rA with respect to X
instead of C

A Different Way to Think About a CSTR

The original design equation is written in terms of outlet flow rate. After introducing conversion, the same reactor can be described in terms of reactor performance. This is usually much more intuitive.

Instead of asking:

What is the outlet flow rate?

we ask:

What conversion does this reactor achieve?

<u>Batch reactor:</u>

$\frac{dN_A}{dt} = r_A V$ and $X = \frac{N_{A0} - N_A}{N_{A0}}$.

Write N_A in terms of X is

$$N_A = N_{AO}(1 - X)$$

Hence the mole balance in terms of X is:

$$\frac{dN_A}{dt} = r_A V \rightarrow \frac{d}{dt}[N_{AO}(1 - X)] = r_A V$$

$$\underbrace{\frac{d}{dt}N_{AO}}_{0} - \frac{d}{dt}(N_{AO}X) = -N_{AO}\frac{dX}{dt} = r_A V$$

final form:

$$\frac{dX}{dt} = \frac{-r_A V}{N_{AO}}$$

if I use this form
I need to write
-rA with respect to X
instead of C

Go on to the next page

Conversion Changes with Time

In a batch reactor:

- Time starts at zero.
- Conversion starts at zero.
- Conversion increases as reaction proceeds.

The design question becomes: How much time is needed to reach a desired conversion? This is the central question of batch-reactor design.

PFR:

$\frac{dF_A}{dV} = r_A$ and $X = \frac{F_{A0}-F_A}{F_{A0}}$

F_A in terms of X is $F_A = F_{A0}(1-X)$. Prove that the mole balance in terms of X is: $F_{A0}\frac{dX}{dV} = -r_A$

$$\frac{dF_A}{dV} = r_A \qquad \frac{d}{dV}(F_{AO}(1-X)) = r_A$$

$$\underbrace{\frac{d}{dV}F_{AO}}_{0} - \frac{d}{dV}(F_{AO}X) = r_A$$

$$-F_{AO}\frac{dX}{dV} = r_A \rightarrow \boxed{\frac{dX}{dV} = \frac{-r_A}{F_{AO}}}$$

Conversion Changes with Position

In a PFR:

- Material enters at $X = 0$.
- Conversion increases as fluid moves through the reactor.
- Different locations have different conversion values.

The design question becomes: How much reactor volume is required to reach a desired conversion?

Summary

Today we introduced conversion as a measure of reactor performance. We learned:

- Flow reactors define conversion relative to the amount fed.
- Batch reactors define conversion relative to the amount initially present.
- Mole balances can be rewritten in terms of conversion.
- Conversion can reduce reactor design problems to a single unknown.

Key Takeaways

☑ **Conversion measures reactor progress.** $X = \text{fraction of reactant consumed}$

☑ **The physical meaning of conversion is the same in every reactor.** Only the accounting changes.

☑ **Conversion is often easier to interpret than number of moles or molar flow rates.**

Engineers typically care more about:

How much reacted?

than

How many moles remain?

☑ **Many reactor design equations in this course will be written in terms of conversion.** Conversion is one of the central variables of chemical reaction engineering.

☑ **Today's most important idea:** Conversion allows us to transform reactor design problems into simpler problems with fewer unknowns.

Summary of mole balances:

	in terms of F or N	in terms of X
CSTR	$V = \frac{F_{AO} - F_A}{-r_A}$	$V = \frac{F_{LR,O} X}{-r_{LR}}$
batch	$\frac{dN_A}{dt} = r_A V$	$\frac{dX}{dt} = \frac{-r_{LR} V}{N_{LR,O}}$
PFR	$\frac{dF_A}{dV} = r_A$	$\frac{dX}{dt} = \frac{-r_{LR}}{F_{LR,O}}$